

AI Agent Research Compendium — Volume 1: Multi-Agent Orchestration

A curated collection of the most impactful and working research papers on multi-agent AI systems, published 2026.

Filtered for quality, practical applicability, and production relevance. Source: VoltAgent/awesome-ai-agent-papers + supplementary repos.

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1. Multi-Agent Coordination & Communication

CORAL: Towards Autonomous Multi-Agent Evolution for Open-Ended Discovery

- **arXiv:** 2604.01658
- **PDF:** <https://arxiv.org/pdf/2604.01658>
- **Key Insight:** Long-running multi-agent systems that self-evolve via shared persistent memory, asynchronous execution, and heartbeat-based interventions achieve 3–10× higher improvement rates than fixed evolutionary-search baselines on 10 math/algorithmic/systems tasks.

DyTopo: Dynamic Topology Routing for Multi-Agent Reasoning via Semantic Matching

- **arXiv:** 2602.06039
- **PDF:** <https://arxiv.org/pdf/2602.06039v1>

- **Key Insight:** Dynamically rewiring agent-to-agent connections at each reasoning round via semantic matching instead of fixed communication topologies improves reasoning quality.

CommCP: Efficient Multi-Agent Coordination via LLM-Based Communication with Conformal Prediction

- **arXiv:** 2602.06038
- **PDF:** <https://arxiv.org/pdf/2602.06038v1>
- **Key Insight:** Conformal prediction filters noisy inter-agent messages to improve multi-robot coordination — practical for production systems where communication reliability matters.

Beyond Rule-Based Workflows: An Information-Flow-Orchestrated Multi-Agents Paradigm via A2A Communication from CORAL

- **arXiv:** 2601.09883
- **PDF:** <https://arxiv.org/pdf/2601.09883v1>
- **Key Insight:** Replaces predefined multi-agent workflows with a dynamic information-flow orchestrator that coordinates agents through natural-language A2A communication — critical for flexible production systems.

The Orchestration of Multi-Agent Systems: Architectures, Protocols, and Enterprise Adoption

- **arXiv:** 2601.13671
- **PDF:** <https://arxiv.org/pdf/2601.13671v1>
- **Key Insight:** Formalizes a unified architectural framework for orchestrated multi-agent systems integrating MCP for tool access and Agent2Agent protocol for peer coordination. **Must-read for production architecture.**

ROMA: Recursive Open Meta-Agent Framework for Long-Horizon Multi-Agent Systems

- **arXiv:** 2602.01848
- **PDF:** <https://arxiv.org/pdf/2602.01848v1>
- **Key Insight:** Breaking large tasks into subtask trees running in parallel across multiple agents handles long-horizon workflows without exceeding context windows.

Epistemic Context Learning: Building Trust the Right Way in LLM-Based Multi-Agent Systems

- **arXiv:** 2601.21742
- **PDF:** <https://arxiv.org/pdf/2601.21742v1>
- **Key Insight:** Builds peer reliability profiles from interaction history so agents learn which peers to trust when uncertain — essential for reliable multi-agent production systems.

When Single-Agent with Skills Replace Multi-Agent Systems and When They Fail

- **arXiv:** 2601.04748
- **PDF:** <https://arxiv.org/pdf/2601.04748v2>
- **Key Insight:** Investigates when a single agent with a skill library can replace multi-agent systems, studying scaling limits and phase transitions in skill selection. **Critical for architecture decisions.**

Phase Transition for Budgeted Multi-Agent Synergy

- **arXiv:** 2601.17311
- **PDF:** <https://arxiv.org/pdf/2601.17311v1>
- **Key Insight:** Theory for predicting when budgeted multi-agent LLM systems improve, saturate, or collapse based on context windows, communication fidelity, and shared-error correlation.

2. Multi-Agent Orchestration Frameworks

MAS-Orchestra: Understanding and Improving Multi-Agent Reasoning Through Holistic Orchestration

- **arXiv:** 2601.14652
- **PDF:** <https://arxiv.org/pdf/2601.14652v2>
- **Key Insight:** Formulates multi-agent orchestration as function-calling reinforcement learning with holistic system-level reasoning and introduces MASBENCH for controlled evaluation.

CTHA: Constrained Temporal Hierarchical Architecture for Stable Multi-Agent LLM Systems

- **arXiv:** 2601.10738
- **PDF:** <https://arxiv.org/pdf/2601.10738v1>
- **Key Insight:** Constrained temporal hierarchical architecture that projects inter-layer communication onto structured manifolds with typed message contracts and authority bounds — production stability patterns.

SYMPHONY: Synergistic Multi-agent Planning with Heterogeneous Language Model Assembly

- **arXiv:** 2601.22623
- **PDF:** <https://arxiv.org/pdf/2601.22623v1>
- **Key Insight:** Using a pool of different LLM agents within MCTS planning increases rollout diversity and improves multi-step reasoning.

StackPlanner: Centralized Hierarchical Multi-Agent System with Task-Experience Memory Management

- **arXiv:** 2601.05890
- **PDF:** <https://arxiv.org/pdf/2601.05890v1>
- **Key Insight:** Hierarchical multi-agent framework that decouples high-level coordination from subtask execution with active task-level memory control and RL-driven experience reuse.

Learning Latency-Aware Orchestration for Parallel Multi-Agent Systems

- **arXiv:** 2601.10560
- **PDF:** <https://arxiv.org/pdf/2601.10560v1>
- **Key Insight:** Latency-aware orchestration that explicitly optimizes the critical execution path under parallel execution — essential for real-time production requirements.

MegaFlow: Large-Scale Distributed Orchestration System for the Agentic Era

- **arXiv:** 2601.07526
- **PDF:** <https://arxiv.org/pdf/2601.07526v2>
- **Key Insight:** Decouples agent training into independent Model, Agent, and Environment services for scheduling tens of thousands of concurrent agent tasks. **Critical for enterprise scale.**

Do We Always Need Query-Level Workflows? Rethinking Agentic Workflow Generation

- **arXiv:** 2601.11147
- **PDF:** <https://arxiv.org/pdf/2601.11147v1>
- **Key Insight:** Examines whether query-level workflow generation is always necessary, proposing a low-cost task-level framework that uses self-prediction with few-shot calibration.

ResMAS: Resilience Optimization in LLM-based Multi-Agent Systems

- **arXiv:** 2601.04694
 - **PDF:** <https://arxiv.org/pdf/2601.04694v1>
 - **Key Insight:** Two-stage framework for enhancing multi-agent system resilience through RL-based topology generation and topology-aware prompt optimization under perturbations.
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3. Multi-Agent Debate & Reasoning

Demystifying Multi-Agent Debate: The Role of Confidence and Diversity

- **arXiv:** 2601.19921
- **PDF:** <https://arxiv.org/pdf/2601.19921v1>
- **Key Insight:** Diversity-aware initialization and confidence-modulated updates improve multi-agent debate, connecting findings from human deliberation research.

DynaDebate: Breaking Homogeneity in Multi-Agent Debate with Dynamic Path Generation

- **arXiv:** 2601.05746
- **PDF:** <https://arxiv.org/pdf/2601.05746v1>
- **Key Insight:** Allocates diverse solution paths to agents, shifts focus to step-by-step logic critique, and uses a trigger-based verification agent to resolve deadlocks.

Dynamic Role Assignment for Multi-Agent Debate

- **arXiv:** 2601.17152

- **PDF:** <https://arxiv.org/pdf/2601.17152v1>
- **Key Insight:** Meta-debate framework dynamically assigns roles by matching model capabilities to positions through proposal and peer review stages.

Mixture-of-Models: Unifying Heterogeneous Agents via N-Way Self-Evaluating Deliberation

- **arXiv:** 2601.16863
- **PDF:** <https://arxiv.org/pdf/2601.16863v1>
- **Key Insight:** Runtime Mixture-of-Models architecture with dynamic expertise broker and quadratic voting consensus enabling small model ensembles to match frontier performance.

ORCH: Many Analyses, One Merge — A Deterministic Multi-Agent Orchestrator

- **arXiv:** 2602.01797
 - **PDF:** <https://arxiv.org/pdf/2602.01797v1>
 - **Key Insight:** Deterministic multi-agent orchestrator where multiple LLMs analyze independently and a merge agent selects the best answer without training.
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4. Multi-Agent Learning & Optimization

Scaling Multiagent Systems with Process Rewards

- **arXiv:** 2601.23228
- **PDF:** <https://arxiv.org/pdf/2601.23228v2>
- **Key Insight:** Per-action process rewards from AI feedback improve credit assignment and sample efficiency when finetuning multi-agent LLM systems.

MonoScale: Scaling Multi-Agent System with Monotonic Improvement

- **arXiv:** 2601.23219
- **PDF:** <https://arxiv.org/pdf/2601.23219v1>
- **Key Insight:** Framework for safely growing multi-agent pools with guaranteed non-decreasing performance across onboarding rounds.

Learning Decentralized LLM Collaboration with Multi-Agent Actor Critic

- **arXiv:** 2601.21972
- **PDF:** <https://arxiv.org/pdf/2601.21972v2>
- **Key Insight:** Actor-critic RL methods for training decentralized LLM agent collaboration across writing, coding, and game-playing tasks.

MARO: Learning Stronger Reasoning from Social Interaction

- **arXiv:** 2601.12323
- **PDF:** <https://arxiv.org/pdf/2601.12323v2>
- **Key Insight:** Decomposes multi-agent social interaction outcomes into per-behavior learning signals to improve LLM reasoning through simulated social environments.

OptimAI: Optimization from Natural Language Using LLM-Powered AI Agents

- **arXiv:** 2504.16918
 - **PDF:** <https://arxiv.org/pdf/2504.16918>
 - **Key Insight:** Four-agent pipeline (Formulator, Planner, Coder, Critic) with UCB bandit scheduling. Mixing Gemma 27B and Llama 70B beats any single model; 88% on NLP4LP vs. 72% prior best.
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5. Multi-Agent Evaluation & Benchmarking

A Large-Scale Study on the Development and Issues of Multi-Agent AI Systems

- **arXiv:** 2601.07136
- **PDF:** <https://arxiv.org/pdf/2601.07136v1>
- **Key Insight:** Analyzes over 42K commits and 4.7K resolved issues across eight leading multi-agent AI systems (LangChain, CrewAI, AutoGen, etc.) for development patterns and ecosystem maturity.

Multi-Agent Teams Hold Experts Back

- **arXiv:** 2602.01011
- **PDF:** <https://arxiv.org/pdf/2602.01011v3>

- **Key Insight:** Self-organizing LLM agent teams cannot always match their best member's performance — important for deciding when to use multi-agent vs. single agent.

When Numbers Start Talking: Implicit Numerical Coordination Among LLM-Based Agents

- **arXiv:** 2601.03846
- **PDF:** <https://arxiv.org/pdf/2601.03846v1>
- **Key Insight:** Investigates covert communication in LLM multi-agent systems through game-theoretic analysis of implicit coordination signals.

LSTM-MAS: A Long Short-Term Memory Inspired Multi-Agent System for Long-Context Understanding

- **arXiv:** 2601.11913
- **PDF:** <https://arxiv.org/pdf/2601.11913v1>
- **Key Insight:** LSTM-inspired multi-agent architecture with worker, filter, judge, and manager agents that emulate gated memory mechanisms for long-context understanding.

Learning Latent Invariant Solution Structure via Multi-Agent Constraint Factorization

- **arXiv:** 2601.15077
- **PDF:** <https://arxiv.org/pdf/2601.15077v1>
- **Key Insight:** Formalizes through operator theory why multi-agent LLM systems access invariant solutions that single agents cannot reach.

This volume covers 26 papers from the Multi-Agent domain. Each paper was selected based on: (1) provides working implementations or reproducible results, (2) addresses production-grade concerns, (3) introduces novel architectural patterns, or (4) provides essential empirical evidence for multi-agent design decisions.

Source: VoltAgent/awesome-ai-agent-papers (363+ papers total, filtered to top ~26 per category)